

Subject Description Form

Subject Code	EE4014 / EE4014B
Subject Title	Intelligent Systems Applications in Electrical Engineering
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	To introduce students to the core principles of intelligent systems and the applications of artificial intelligence in Electrical Engineering.
Subject Intended Learning Outcomes	Upon completion of the subject, students will: - Gain a solid understanding of the fundamental concepts, characteristics, methodologies and usefulness of artificial intelligence algorithms and intelligent systems. - Acquire the ability to understand and apply various intelligent system techniques, such as neural networks, supervised learning, unsupervised learning, reinforcement learning, generative learning, computer vision, large language model, and evolutionary computation. - Achieve strong programming skills, including the ability to design, implement, and debug AI algorithms using appropriate programming languages and frameworks (e.g., Python, TensorFlow, PyTorch). - Develop the capability to integrate intelligent system approaches into real-life problem-solving - Develop skills in presenting and interpreting results obtained from the mini-project, and effectively communicating them in written form.
Subject Synopsis/ Indicative Syllabus	Artificial neural network: Concepts of intelligent systems. Neuron and perceptron. Multi-layer neural network. Supervised learning. Forward and backward propagation. Training of neural networks. Recurrent and convolutional neural network. Unsupervised learning: Concepts. K-means. Competitive learning and self-organizing map. Deep and Generative Learning: Deep neural network. Reinforcement learning. Generative learning. Attention mechanism and large language model. Evolutionary computation: Concepts. Genetic algorithm. Particle swarm optimization. Applications of intelligent systems in practical systems. Mini-project: Utilize the introduced intelligent system techniques to solve engineering problems.

Teaching/Learning Methodology	Lectures and tutorials serve as the primary methods for conveying fundamental concepts and theories. Practical experiences in system analysis, design, and applications are provided through the mini-project. In the project, students are expected to solve engineering problems using AI techniques, employing critical and analytical thinking. The mini-project is designed to complement the lecture materials, encouraging students to engage in additional readings and seek out relevant information.																																																					
	Teaching/Learning Methodology		Outcomes																																																			
			a	b	c	d	e																																															
	Lectures		✓	✓	✓	✓																																																
	Tutorials		✓	✓	✓	✓																																																
	Mini-projects				✓	✓	✓																																															
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><td rowspan="2">Specific assessment methods/tasks</td><td rowspan="2">% weighting</td><td colspan="5">Intended subject learning outcomes to be assessed</td></tr><tr><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td></tr><tr><td>1. Examination</td><td>60%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>2. Class Test</td><td>18%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>3. Mini-project/report</td><td>12%</td><td></td><td></td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>4. Cases study & presentation exercises</td><td>10%</td><td></td><td></td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100%</td><td colspan="4"></td><td></td></tr></table> The outcomes of concepts, design, and applications are assessed by the usual means of examination and class test. Mini-project and cases study & presentation assess those on programming skills, problem-solving techniques and practical considerations of intelligent system applications, as well as technical reporting, teamwork and presentation skills.							Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed					a	b	c	d	e	1. Examination	60%	✓	✓	✓			2. Class Test	18%	✓	✓	✓			3. Mini-project/report	12%			✓	✓	✓	4. Cases study & presentation exercises	10%			✓	✓	✓	Total	100%					
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Student Study Effort Expected	Class contact:																																																					
	▪ Lecture/Tutorial						33 Hrs.																																															
	▪ Cases study & presentation						6 Hrs.																																															
	Other student study effort:																																																					
	▪ Mini-project and report						16 Hrs.																																															
	▪ Self-study						50 Hrs.																																															
	Total student study effort						105 Hrs.																																															

Reading List and References	Reference books: 1. Management Association, Information Resources, ed. Deep Learning and Neural Networks: Concepts, Methodologies, Tools, and Applications. Hershey, PA: IGI Global, 2020 2. E. Alpaydin, Machine Learning, The MIT Press, 2021 3. A. Ye, Modern Deep Learning Design and Application Development: Versatile Tools to Solve Deep Learning Problems, Apress, 2022 4. M. Negnevitsky, Artificial Intelligence - A Guide to Intelligent Systems, Addison-Wesley, 2011 5. K.Y. Lee and M.A. El-Sharkawi, Modern Heuristic Optimization Techniques: Theory and Applications to Power Systems, Wiley-IEEE Press, 2008 6. Articles from IEEE Transactions on Artificial Intelligence
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